Biology Class 11 National Book Foundation FBISE Quality Notes
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Chapter 2: Biological Molecules

SLO 1: Biochemistry / Molecular biology (2.1)

Biochemistry: *The study of the structure of different chemical compounds found in living organisms and the chemical processes taking place within or related to living organisms.*

- *It is a sub-discipline of both biology and chemistry. Molecular biology is a specialized branch of biochemistry.*
- *Importance: explains cell biology, anatomy, physiology, evolution and variation in life forms in chemical terms (e.g. muscle contraction).*

SLO 2: Bonds and interactions in biological molecules (2.2)

Bond / interaction	Details and examples
Covalent bond	Mutual sharing of electrons between two atoms. Most prevalent (C, H, O, N, S, P are non-metals). C-C, C-H, peptide, glycosidic, phosphodiester, disulphide, ester and thioester bonds.
Ionic bond	One atom donates electron(s) and becomes positive ion, the other accepts and becomes negative; held by electrostatic attraction. Less common; in salts, between some amino acids in proteins, phosphate groups of DNA, metal ions and enzymes.
Hydrogen bond	H attached to a highly electronegative atom (O, N, F) attracted to a lone pair of a neighbouring electronegative atom. DNA double helix: 3 H-bonds between C and G, 2 between A and T.
Hydrophobic interaction	"Afraid of water": non-polar molecules cluster together in water. Protein folding, membrane formation, ligand binding.

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Bond / interaction	Details and examples
Hydrophilic interaction	"Water-loving": polar molecules (partial + and - poles). Hydration of ions; phospholipid heads outward; DNA/RNA phosphates dissolve in water; enzyme active sites.

SLO 3: Protoplasm and fundamental biomolecules (2.3)

Protoplasm: Living mass of the cell; all biochemical reactions occur in it. It is made of bioelements and biomolecules.

- Out of 92 natural elements about **25** are found in living organisms (**bioelements**); human body has only **16**. Only **six** major bioelements make about 99% of protoplasm; minor bioelements are less than 1% (e.g. Cl 0.15%, Na 0.15%, Mg 0.05%).
- Biomolecules: inorganic (water, minerals) and organic (carbohydrates, lipids, proteins, nucleic acids).
- **Four fundamental biomolecules** are present in every cell, from simplest prokaryote to mammalian cell:

Biomolecule	Role in the cell
Carbohydrates	Fuel for metabolic activities.
Proteins	Membranes, ribosomes, cytoskeleton, enzymes.
Lipids	Membranes and cytoplasm; reserve energy, shape, protect and insulate.
DNA / RNA	DNA in chromosomes controls cell activity; RNA carries genetic information and helps protein synthesis.

SLO 4: Condensation and hydrolysis (2.4)

- **Macromolecule (polymer):** high molecular weight compound made of many repeating units; the units are **monomers**.

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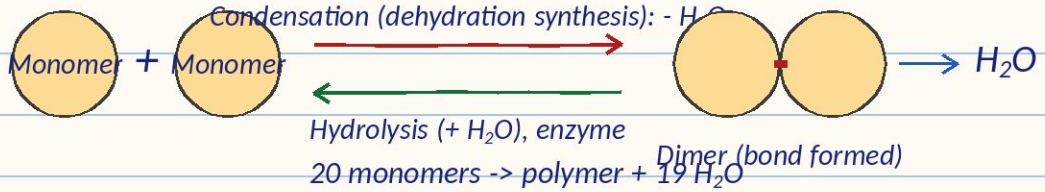
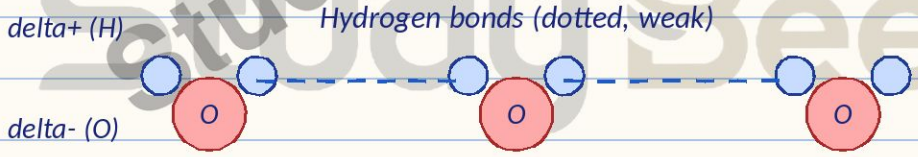


Fig: Condensation and hydrolysis

- ***Condensation (dehydration synthesis):*** *-OH of one monomer and -H of the other are removed as water and a bond forms. Two monomers give a dimer (e.g. two alpha-glucose = maltose). Repeated = polymer (amylose). 20 monomers release 19 water molecules. Needs enzyme and activated (energy rich) monomers.*
- ***Hydrolysis:*** *reverse of condensation; breakdown of dimer/polymer by adding water (-OH to one monomer, -H to other). All digestion reactions are hydrolysis (carbohydrases, proteases, lipases, nucleases).*

SLO 5: Water as medium of life (2.5)

- *About 70% of any organism is water; lowest 20% in seeds and bones, highest 85-90% in brain cells, jellyfish 99%.*



H_2O = polar molecule: O slightly negative, each H slightly positive

Fig: Polarity and hydrogen bonding of water

Property	***Explanation and importance***
1. High polarity	*Unequal sharing of electrons: O slightly negative, H slightly positive. Makes water the universal solvent; all reactions occur in aqueous medium.*
2. Hydrogen bonding	*Weak bonds between molecules (dotted lines). Water is liquid at temperatures suitable for life.*

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Property	Explanation and importance
3. Cohesion and adhesion	Cohesion = attraction among water molecules; adhesion = attraction to polar surfaces. Water flows and acts as transport medium.
4. High specific heat	1 calorie (4.18 J) raises 1 g by 1 degree C. Much heat is used to break H-bonds, so temperature rises little. Temperature stabilizer.
5. High heat of vapourization	574 calories per gram. Sweating cools animals; evaporation of 2 ml of 1 litre lowers temperature of remaining 998 ml by 1 degree C.
6. Hydrophobic exclusion	Reduces contact between water and hydrophobic substances (oil drops join). Keeps lipid bilayer membrane intact.
7. Ionization	Water dissociates into H^+ and OH^- (reversible, equilibrium at 25 C). Helps maintain or change pH.
8. Lower density of ice	Ice has lattice of max H-bonds, floats. Surface ice insulates the water below so organisms live under it.

SLO 6: Carbohydrates - definition and classification (2.6)

- Compounds of C, H, O; "hydrates of carbon" with formula $C_n(H_2O)_n$. Glucose = $C_6H_{12}O_6$. Not exact for all (lactic acid $C_3H_6O_3$ is not a carbohydrate; deoxyribose $C_5H_{10}O_4$ is).

Carbohydrates: Polyhydroxy aldehydes or polyhydroxy ketones (monosaccharides), or their complex derivatives which on hydrolysis give polyhydroxy aldehydes or ketones.

Monosaccharides	Oligosaccharides	Polysaccharides
Single saccharide unit; cannot be hydrolysed	2 to 10 units	More than 10 units
Highly soluble in water	Less soluble	Generally insoluble
Sweetest	Less sweet	Tasteless

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SLO 7: Monosaccharides (2.7)

- General formula $(CH_2O)_n$; 3 to 7 carbons. All carbons except one carry -OH; the remaining carbon is part of aldehyde (**aldoses**) or ketone (**ketoses**).

Type	Examples and role
Trioses (3C)	Glyceraldehyde, dihydroxyacetone: intermediates in photosynthesis and respiration.
Tetroses (4C)	Erythrose.
Pentoses (5C)	Ribose, deoxyribose (nucleic acids); ribulose (photosynthesis).
Hexoses (6C)	Glucose (respiratory substrate), fructose (intermediate in respiration), galactose (in milk sugar).
Heptoses (7C)	Glucoheptose.

Properties of monosaccharides

1. Simplest carbohydrates: cannot be hydrolysed further.
2. Highly soluble: many -OH groups form H-bonds with water.
3. Sweet taste: arranged -OH groups activate sweet taste receptors.
4. **Reducing sugars:** free aldehyde or ketone group donates electrons.
5. Show **stereoisomerism** (asymmetric carbon = attached to four different groups).
6. Five or more carbons exist as open chain or ring forms.

Ring structures

- Open chain in crystalline form; in water pentoses and hexoses form rings. **Furanose:** 5-membered ring (1 O + 4 C; C1 and C4 joined through O); all pentoses and ketohexoses. **Pyranose:** 6-membered ring (1 O + 5 C; C1 and C5); only aldohexoses e.g. glucopyranose. Ribose forms ribofuranose.

SLO 8: Isomers and stereoisomers of glucose (2.8)

- **Isomerism:** same molecular formula, different structure. Glucose (aldose) and fructose (ketose) both $C_6H_{12}O_6$ = functional group isomers.
- **Stereoisomers:** same formula, differ in arrangement of -H and -OH in 3D space. Number = 2^n (n = asymmetric carbons). Glucose has 4 asymmetric

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carbons (C2-C5) so **16** stereoisomers. Anomeric carbon = C1 of ring (carbonyl carbon); penultimate carbon = C5.

Glucopyranose ring: O links C1 and C5 (pyranose = 6-membered)

Fig: Anomers of glucose

Type	Meaning / example
Anomers	Differ at C1 only: alpha (-OH downward) and beta (-OH upward).
Enantiomers	Non-superimposable mirror images: D (-OH right at C5) and L (-OH left). 8 of the 16 are enantiomers of the other 8.
Diastereoisomers	Differ at more than one asymmetric carbon; not mirror images, e.g. D-glucose and D-altrose.
Epimers	Differ at only one asymmetric carbon, e.g. D-glucose and D-mannose.

- Living organisms use only D-sugars; enzyme active sites fit D-substrates, so L-sugars (tagatose, sucralose) cannot be metabolized.

SLO 9: Disaccharides and glycosidic bonds (2.9, 2.10)

- Two monosaccharides joined; formula $C_{12}H_{22}O_{11}$; less sweet and less soluble; hydrolysed to monosaccharides; energy source.

Disaccharide	Made of / bond / role
Sucrose (cane sugar)	alpha-glucose + beta-fructose; alpha-1,2 glycosidic bond. Transport sugar in plants (soluble, unreactive). Non-reducing.
Maltose (malt sugar)	alpha-glucose + alpha-glucose; alpha-1,4 bond (C1 -OH with H of C4 -OH). Product of starch / glycogen breakdown; in germinating seeds, brewing (amylase on barley).

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Disaccharide	Made of / bond / role
Lactose (milk sugar)	beta-galactose + beta-glucose; beta-1,4 bond. In milk: goat 3-4%, cow 4-6%, human 5-8%. Energy for infants.

Glycosidic bond: Covalent bond between two sugar (ringed) molecules formed by condensation, with release of one water molecule. Present in di-, oligo- and polysaccharides. Named by carbon numbers and alpha/beta type.

- Maltose and lactose have a free open-chain group (reducing); sucrose and polysaccharides are non-reducing.

SLO 10: Polysaccharides (2.11)

- Over 10 monosaccharide units. **Homopolysaccharides** (one kind): starch, glycogen, cellulose, chitin. **Heteropolysaccharides** (different kinds): agar, pectin, peptidoglycan.
- Good storage molecules: large and insoluble (no osmotic effect), compact, easily hydrolysed to sugars.

Polysaccharide	Structure and role
Starch	Hundreds of alpha-glucose; storage of plants (root, stem, seeds). Amylose: unbranched, alpha-1,4, soluble in hot water. Amylopectin: branched by alpha-1,6, insoluble. Digested by amylase to maltose, then maltase to glucose. Iodine gives blue-black.
Glycogen	Alpha-glucose; animal starch, stored in liver and muscles; like amylopectin but much more branched. Iodine gives red colour.
Cellulose	Most abundant carbohydrate; beta-glucose with beta-1,4 bonds, unbranched; plant cell wall; cotton and paper. No colour with iodine. Not digested by humans (roughage/fibre); herbivores use bacteria.
Chitin	2nd most abundant organic molecule; N-acetyl glucosamine monomers, beta-1,4, unbranched; fungal cell wall (fungal cellulose) and arthropod exoskeleton.

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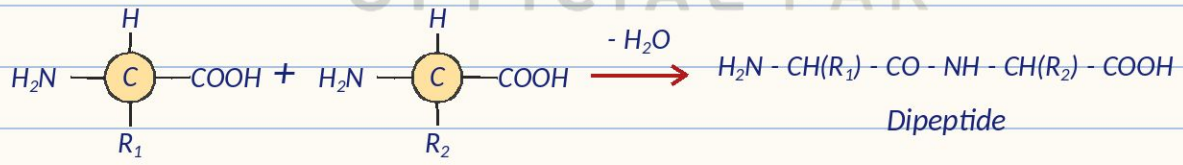
Biochemical tests

Test	Method and result
Iodine test (starch)	Add iodine drops: dark blue-black = starch; red = glycogen.
Fehling / Benedict (reducing sugar)	Fehling solution + glucose solution, heat: coloured precipitate = reducing sugar.
Molisch / alpha-naphthol (sucrose)	Add alcoholic alpha-naphthol, pour conc. H_2SO_4: purple ring at junction.

SLO 11: Proteins and amino acids (2.12)

- Proteins: polymers of amino acids (polypeptide chains); contain C, H, O, N, some P, S; few have Fe, I, Mg.
- **Amino acid:** alpha carbon + H + amino group ($-NH_2$) + carboxyl group (-COOH) + variable R group. Only **20** are common in proteins; R decides the properties. Simplest: glycine and alanine.
- **Essential amino acids (9):** cannot be made by body, must come from diet: lysine, leucine, isoleucine, histidine, methionine, phenylalanine, threonine, valine, tryptophan.
- **Non-essential (11):** made by body: alanine, arginine, asparagine, aspartic acid, cysteine, glutamic acid, glutamine, glycine, proline, serine, tyrosine.

SLO 12: Peptide linkage (2.13)



CO - NH = peptide bond (C of -COOH to N of $-NH_2$)

General amino acid: alpha C + H + amino ($-NH_2$) + carboxyl (-COOH) + R group

Fig: Amino acids and formation of peptide bond

- -OH of the carboxyl group of one amino acid and -H of the amino group of another are lost as water; C-N **peptide bond** forms = dipeptide. Occurs at

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*ribosome under mRNA instructions (translation). Dipeptide has **N-terminal** (amino) and **C-terminal** (carboxyl) ends; more amino acids added at C-end = tri-, polypeptide.*

- *Peptide bonds are broken by hydrolysis: pepsin, trypsin, erepsin in gut; lysosomal enzymes inside cells.*

Levels of protein structure

Level	Description
Primary	Linear sequence and number of amino acids; shown by all proteins at synthesis; held by peptide bonds.
Secondary	Alpha-helix or beta-pleated sheet; H-bonds between groups of different amino acids.
Tertiary	Helix folds again and again into globular shape by ionic bonds and disulphide bridges.
Quaternary	More than one globule (polypeptide) joined by hydrophobic interactions.

SLO 13: Significance of amino acid sequence - sickle cell (2.14)

- *The sequence is determined by the nucleotide sequence of DNA; a point mutation can change it and cause severe defects.*
- *Normal haemoglobin (HbA): 4 polypeptides - two alpha chains (141 amino acids each) and two beta chains (146 each). Carries oxygen; normal RBCs are disc shaped.*
- ***Sickle cell anaemia:** hereditary; point mutation in beta-globin gene (one nucleotide replaced). In HbS, **glutamic acid is replaced by valine at position 6** of the beta chain. Cells become stiff, sticky, crescent shaped, block blood vessels: pain and organ damage.*

SLO 14: Classification and roles of proteins (2.15, 2.16)

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Fibrous proteins	Globular proteins
Fibre / filament shape	Spherical shape
Secondary structure during function	Tertiary or quaternary structure
Insoluble in water, elastic, cannot be crystallized	Soluble, inelastic, can be crystallized
Collagen, fibrinogen, actin, myosin, keratin	Enzymes, hormones, antibodies, channel proteins

Structural proteins	Role
Collagen	Matrix of bone and cartilage; ligaments and tendons.
Elastin	Support for connective tissue (tendons, ligaments).
Keratin	Hair, nails, quills, feathers, horns, beaks.
Histones	Arrange DNA into chromosome.
Functional proteins	**Role**
Enzymes	Speed up biochemical reactions (control metabolism).
Hormones	Regulate glucose, calcium, digestion, blood pressure.
Antibodies	Made by WBCs against antigens; immunity.
Haemoglobin	In RBCs; transports oxygen, some CO_2.
Fibrinogen	In plasma; blood clotting.
Ovalbumin, casein	Egg white and milk proteins; storage of amino acids.

Tests for protein

Test	Result
Biuret ($NaOH + 1\% CuSO_4$)	Pink or violet colour = protein.
Millon's reagent	White precipitate turns red on heating.
Ninhydrin (boil)	Coloured product = amino acids / protein present.

SLO 15: Lipids (2.17)

Lipids: Heterogeneous group of organic compounds (fats, oils, waxes, steroids) insoluble in water but soluble in organic solvents (acetone, alcohol, ether).

- C, H, O; less oxygen than carbohydrates, so they give about **double energy**. Tristearin = $C_{57}H_{110}O_6$.

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- *Roles: membranes (phospholipids, cholesterol), energy store (triglycerides), steroid hormones, protection, waterproofing, insulation, buoyancy.*

SLO 16: Acylglycerols, phospholipids, terpenes, waxes (2.18)

Lipid	Properties and roles
Acylglycerols	Less dense than water; insoluble in water, soluble in chloroform, ether; saturated long-chain ones melt higher (tristearin 71 C, triolein -17 C); emulsifying power; hydrolysis gives fatty acids + glycerol. Major lipids of fat deposits; excess causes obesity, diabetes; hydrogenation gives margarine.
Phospholipids	Amphipathic (polar and non-polar ends); fix proteins in bilayer; regulate membrane permeability; fat absorption; ETC in mitochondria; carry cholesterol; surfactant in lungs; blood coagulation.
Terpenes	Anti-inflammatory, analgesic, antioxidant, anticancer, antibiotic, neuroprotective, antidiabetic.
Waxes	Polishing, waterproofing; used in cosmetics, candles, crayons, paints, inks, adhesives, tyres.

SLO 17: Molecular structure of acylglycerol, phospholipid, terpene (2.19)

- ***Acylglycerol:*** *ester of glycerol (3C trihydroxy alcohol) + fatty acids (R-COOH, even number of C, 2 to 30).* ***Esterification*** *releases water. 1 fatty acid = monoacylglycerol; 2 = diacylglycerol; 3 =* ***triacylglycerol*** *(neutral lipid, no free -OH left).*
- ***Fatty acids:*** *acetic (2C), butyric (4C) simplest; palmitic (16C), stearic (18C) most common.* ***Saturated*** *(no double bond, e.g. palmitic): solid at room temperature, animal fats.* ***Unsaturated*** *(one or more double bonds, e.g. oleic): liquid, plant oils.*
- ***Waxes:*** *esters of long chain fatty acids and long chain alcohols; beeswax, cutin on leaves; chemically inert. Synthetic: paraffin wax.*

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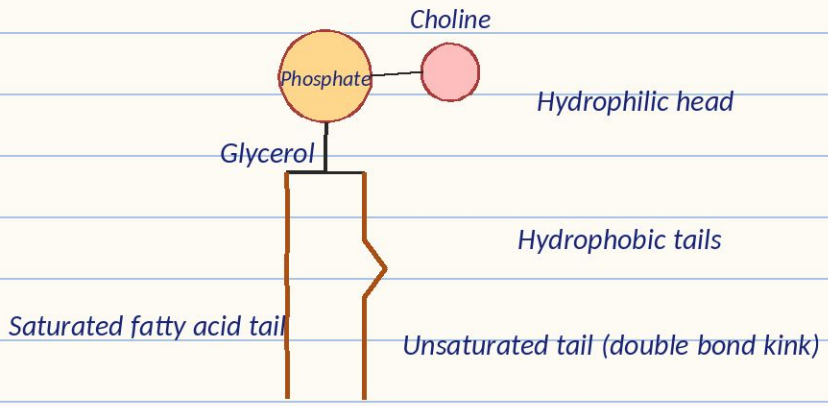

Fig: Structure of a phospholipid (phosphatidylcholine)

- ***Phospholipid:*** *phosphatidic acid = glycerol + 2 fatty acids + phosphate; joined to choline, ethanolamine, inositol or serine. Commonest = phosphatidylcholine (**lecithin**). Head hydrophilic, tails hydrophobic; main part of membrane bilayer.*
- ***Terpenes:*** *made from five-carbon **isoprene** unit. 2 units = monoterpene (menthol); 4 = diterpene (vitamin A, phytol); 6 = triterpene (ambrein); natural rubber = polyterpene.*

SLO 18: Steroids and prostaglandins (2.20)

- ***Steroids:*** *high molecular weight crystalline lipids; nucleus of 17 carbons in four rings (three 6-carbon, one 5-carbon); made from isoprene units. Cholesterol (membrane component) is precursor of bile acids, testosterone, progesterone, estrogen; vitamin D (calcium metabolism) is also a steroid.*
- ***Prostaglandins:*** *local hormones in almost every mammalian tissue, derived from arachidonic acid. Raise or lower blood pressure, induce fever, inflammation and pain, regulate platelet aggregation. **Aspirin** reduces fever and pain by inhibiting prostaglandin synthesis.*

Tests for lipids

- *Spot test: greasy translucent spot on filter paper. Emulsion test: cloudy white suspension on adding cold water to lipid in alcohol. Sudan III: dark red ring/layer at the surface.*

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SLO 19: Nucleic acids and nucleotides (2.21, 2.22)

- *First reported in 1869 by a Swiss physician (from pus cell nuclei), named **nuclein**; renamed nucleic acid (acidic) in 1920. Two types: DNA and RNA; linear unbranched polymers; monomer = **nucleotide**.*

***Nucleotide:** Pentose sugar + phosphate + nitrogenous base. (Deoxyribose in DNA, ribose in RNA.)*

- ***Nucleoside** = base + pentose sugar (base joined to 1' carbon). Phosphate added at 5' carbon gives nucleotide: monophosphate (one), diphosphate, triphosphate.*

Pyrimidines (single ring)	**Purines (double ring)**
Cytosine (C), Thymine (T, only DNA), Uracil (U, only RNA)	Adenine (A), Guanine (G)

SLO 20: ATP and NAD (2.24)

- ***ATP** (mononucleotide): adenine + ribose + three phosphates. The two bonds between phosphates are **high energy bonds**. ATP to ADP + Pi by hydrolysis releases energy; so ATP is the **energy currency of the cell**.*
- ***NAD** (dinucleotide): two nucleotides joined by their phosphates - one with nicotinamide, one with adenine. It is a coenzyme.*

SLO 21: Phosphodiester bond (2.23)

- *Nucleotides join by condensation, but the molecule released is **pyrophosphate** (two phosphates), not water.*

1. *Phosphate at 5' of incoming nucleoside triphosphate (NTP) is held near free 3' -OH of the last nucleotide; **phosphodiester bond** forms.*
2. *Bond between alpha and beta phosphates of NTP breaks; pyrophosphate is released.*
3. *Pyrophosphate is hydrolysed, releasing energy that drives polymerization forward.*

- *A polynucleotide has a free 5' phosphate at one end and free 3' -OH at the*

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other; sequences are named 5' to 3'.

SLO 22: Watson and Crick model of DNA (2.25)

- **Chargaff (1951):** A = T and G = C, so purines : pyrimidines = 1 : 1. X-ray work of Wilkins and Franklin showed DNA is a double helix (width 2 nm, one turn 3.4 nm). Watson and Crick (Cambridge, 1953) proposed the model; Nobel Prize 1962.

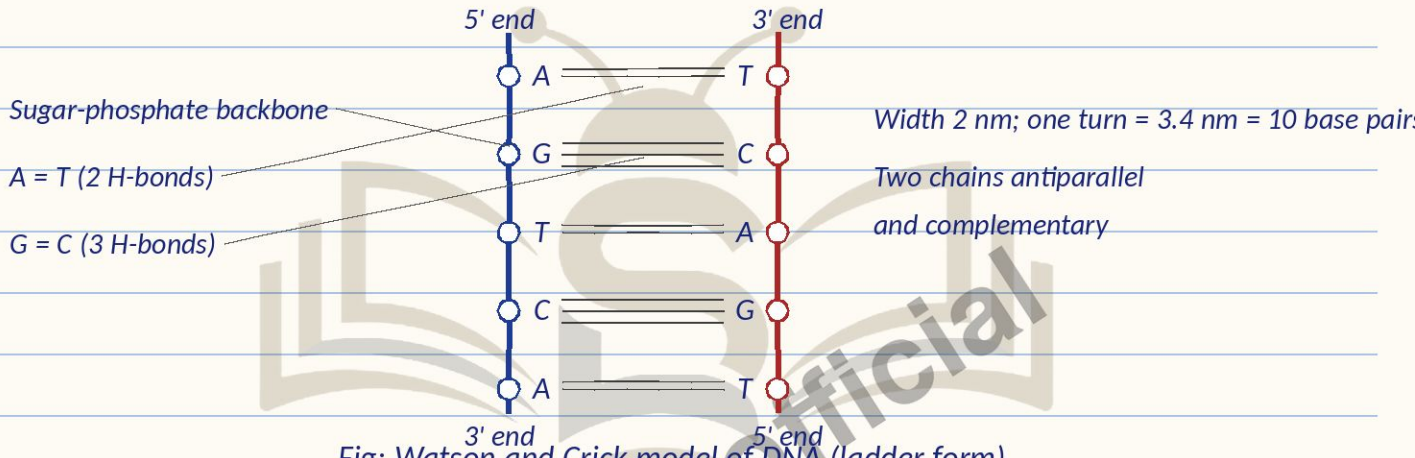

Fig: Watson and Crick model of DNA (ladder form)

- Two polynucleotide chains joined by base pairs; **antiparallel** (one 5' to 3' downward, other upward); constant width 2 nm.
- A pairs with T (2 H-bonds), G pairs with C (3 H-bonds); chains are complementary; 10 base pairs per turn. Sequence can vary endlessly and is specific to species and individuals.

SLO 23: RNA and its types (2.26, 2.27)

- RNA is a polymer of nucleotides; generally **single stranded** (some regions double stranded); transcribed from DNA template. Three major classes:

RNA	Features and function
mRNA	3-4% of total RNA; single strand of variable length. Every 3 nucleotides = a **codon** for one amino acid (100 amino acids need 300 nucleotides). Carries message from nucleus to ribosome for translation.
rRNA	About 80% of total RNA; largest; part of ribosome with proteins; sequence similar from bacteria to humans; platform for protein synthesis.

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RNA	Features and function
tRNA	Smallest (75-90 nucleotides); cloverleaf shape; 5' end G, 3' end CCA where amino acid attaches; middle loop of 7 bases has the **anticodon** complementary to mRNA codon; carries a specific amino acid to ribosome. Human cells: about 45 kinds.

SLO 24: Central dogma (2.28)

Transcription + translation = gene expression

Fig: Central dogma of life

- Flow of information: DNA to RNA to protein.
- **Transcription:** two DNA strands of a gene unwind; mRNA is made by pairing complementary RNA nucleotides (A, C, G, U) with the exposed DNA strand; uracil pairs with adenine; DNA zips back.
- **Translation:** mRNA moves to cytoplasm, binds ribosome; tRNAs with matching anticodons bring amino acids; each codon specifies one amino acid; peptide bonds form polypeptide.
- Why DNA has H at 2' carbon: a 2' -OH (as in ribose) can cleave the phosphodiester bond, so DNA is a stable long-term store; RNA degrades in about 30 minutes.

SLO 25: Conjugated molecules (2.29)

Conjugated molecules: Molecules joined with other kinds of molecules.

Conjugated molecule	Made of / occurrence / role
Glycolipids	Lipid + simple sugars; white matter of brain, myelin sheath, chloroplast membrane.
Glycoproteins	Protein + carbohydrate; hormones, transport proteins, receptors; blood group antigens.

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Conjugated molecule	Made of / occurrence / role
Lipoproteins	Protein + phospholipids; milk, blood, nucleus, egg yolk, chloroplasts.
Nucleoproteins	Basic protein + nucleic acid; chromosomes and ribosomes.

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Chapter 2: Short Questions with Answers (SLO based)

Q1. [SLO 1] Define biochemistry. How is it related to molecular biology?

Ans: Biochemistry is the study of the structure of chemical compounds in living organisms and the chemical processes within them. Molecular biology is a specialized branch of biochemistry.

Q2. [SLO 1] How would you describe the importance of biochemistry?

Ans: It helps to explore cell biology, anatomy and physiology, because structures and activities of living organisms (e.g. muscle contraction) are explained in biochemical terms; evolution and variation are also discussed biochemically.

Q3. [SLO 2] What is a covalent bond? Why is it the most prevalent bond in biomolecules?

Ans: A bond formed by mutual sharing of electrons between two atoms. Biomolecules are mainly made of non-metals (C, H, O, N, S, P), so covalent bonds are most common.

Q4. [SLO 2] Name four covalent bonds found in biological molecules.

Ans: Peptide bond, glycosidic bond, phosphodiester bond, disulphide bond (also ester, thioester, C-C and C-H bonds).

Q5. [SLO 2] What is an ionic bond? Give one role in biological molecules.

Ans: Attraction between oppositely charged ions formed when one atom donates and another accepts electrons. It is found between some amino acids in proteins and in enzyme-substrate interactions.

Q6. [SLO 2] What is the importance of hydrogen bonding in biological molecules?

Ans: It holds the two chains of DNA together (3 H-bonds C-G, 2 H-bonds A-T) and takes part in protein folding and protein-DNA interactions.

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Q7. [SLO 2] Differentiate hydrophobic and hydrophilic interactions.

Ans: Hydrophobic: non-polar molecules cluster together away from water (membranes, protein folding). Hydrophilic: polar molecules interact with water and dissolve (hydration of ions, DNA phosphate backbone).

Q8. [SLO 3] What are bioelements?

Ans: Out of 92 natural elements about 25 are found in living organisms; these are called bioelements. Human body has 16 of them.

Q9. [SLO 3] Differentiate between major and minor bioelements.

Ans: Six major bioelements make about 99% of protoplasm; minor bioelements are less than 1% (e.g. Cl 0.15%, Na 0.15%, Mg 0.05%).

Q10. [SLO 3] Describe the chemical composition of protoplasm.

Ans: Protoplasm consists of bioelements which form inorganic biomolecules (water, minerals) and organic biomolecules (carbohydrates, lipids, proteins, nucleic acids).

Q11. [SLO 3] What are the four fundamental kinds of biological molecules?

Ans: Carbohydrates (fuel), proteins (structure, enzymes), lipids (membranes, energy store) and nucleic acids (DNA controls cell, RNA helps protein synthesis). They are present in every cell.

Q12. [SLO 4] Define monomer, polymer and macromolecule.

Ans: Monomers are individual units; a polymer or macromolecule is a high molecular weight compound made from many repeating monomers.

Q13. [SLO 4] What is condensation? Why is it called dehydration synthesis?

Ans: Joining of monomers by removing -OH from one and -H from other as water, forming a bond. Water is removed (dehydration) and a bond is made (synthesis).

Q14. [SLO 4] How many water molecules are released when 20 monomers condense to a polymer?

Ans: 19 water molecules (one for each bond formed).

Q15. [SLO 4] Define hydrolysis. Give an example.

Ans: Breakdown of a dimer or polymer into monomers by addition of water. Example: amylose to alpha-glucose by amylase; all digestion reactions.

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Q16. [SLO 5] Why is the covalent bond in water polar?

Ans: Electrons are not shared equally between O and H; they are pulled closer to oxygen (more electronegative) making it slightly negative and H slightly positive.

Q17. [SLO 5] Why is water regarded as universal solvent?

Ans: Because of its polarity it dissolves ionic compounds and polar substances; they dissociate into ions and become ready to react, so all reactions of life occur in water.

Q18. [SLO 5] What is the importance of hydrogen bonding in water?

Ans: It keeps water liquid at temperatures suitable for life and gives cohesion and adhesion, making water a transport medium.

Q19. [SLO 5] Define cohesion and adhesion.

Ans: Cohesion: attraction among water molecules. Adhesion: attraction of water to polar surfaces. Both are due to hydrogen bonds.

Q20. [SLO 5] Why does a very large amount of heat raise the temperature of water very little?

Ans: Water has high specific heat; most absorbed heat is used to break hydrogen bonds, so the temperature rises little.

Q21. [SLO 5] How does water protect living things against sudden thermal change?

Ans: Its high heat capacity makes it a temperature stabilizer / regulator for organisms in hot environments.

Q22. [SLO 5] What is the importance of high heat of vapourization of water to animals?

Ans: Water has 574 calories per gram heat of vapourization, so evaporation of a little sweat removes much body heat and cools the animal.

Q23. [SLO 5] What is hydrophobic exclusion? Give its biological role.

Ans: Reduction of contact between water and hydrophobic substances (oil drops join into one). It maintains the integrity of lipid bilayer membranes.

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Q24. [SLO 5] Why does ice float on water? What is its importance?

Ans: Ice has a lattice with maximum hydrogen bonding, so it is less dense. The surface ice insulates the water below so aquatic life survives.

Q25. [SLO 5] What is the ionization of water and why is it important?

Ans: Water dissociates reversibly into H^+ and OH^-. These ions take part in cell reactions and help to maintain or change pH.

Q26. [SLO 6] Define carbohydrates. Why is the formula $C_n(H_2O)_n$ not always applicable?

Ans: Carbohydrates are polyhydroxy aldehydes or ketones or their derivatives. The formula fails for lactic acid ($C_3H_6O_3$, not a carbohydrate) while deoxyribose ($C_5H_{10}O_4$) is a carbohydrate.

Q27. [SLO 6] Describe the classification of carbohydrates.

Ans: Based on number of saccharide units: monosaccharides (1 unit), oligosaccharides (2 to 10 units) and polysaccharides (more than 10 units).

Q28. [SLO 7] Describe the classification of monosaccharides.

Ans: By functional group: aldoses (aldehyde) and ketoses (ketone). By carbon number: trioses (3C), tetroses, pentoses, hexoses and heptoses (7C).

Q29. [SLO 7] Why are monosaccharides called reducing sugars?

Ans: They have a free aldehyde or ketone group whose oxygen donates electrons to other compounds, reducing them while they get oxidized.

Q30. [SLO 7] Differentiate between furanose and pyranose.

Ans: Furanose is a five-membered ring (4 C + 1 O; C1-C4 linked via O) of pentoses and ketohexoses. Pyranose is a six-membered ring (5 C + 1 O; C1-C5) of aldohexoses like glucose.

Q31. [SLO 8] Differentiate between isomers and stereoisomers.

Ans: Isomers have the same molecular formula but different structure (glucose and fructose). Stereoisomers differ only in the 3D arrangement of -H and -OH at asymmetric carbons.

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Q32. [SLO 8] What are anomers? How do alpha and beta glucose differ?

Ans: Stereoisomers differing at C1 (anomeric carbon). In alpha the -OH on C1 is downward; in beta it is upward.

Q33. [SLO 8] What are enantiomers? Give an example.

Ans: Non-superimposable mirror images, e.g. D-glucose and L-glucose (-OH right or left on C5).

Q34. [SLO 8] Differentiate between epimers and diastereoisomers.

Ans: Epimers differ at only one asymmetric carbon (D-glucose and D-mannose). Diastereoisomers differ at more than one asymmetric carbon and are not mirror images (D-glucose and D-altrose).

Q35. [SLO 8] Why does glucose have 16 stereoisomers?

Ans: Number = 2^n; glucose has four asymmetric carbons (C2, C3, C4, C5), so $2^4 = 16$.

Q36. [SLO 8] Why can human enzymes not metabolize L-sugars?

Ans: Enzyme active sites are right-handed and fit only D-sugars, so left-handed substrates cannot fit.

Q37. [SLO 9] Write general formula and properties of disaccharides.

Ans: $C_{12}H_{22}O_{11}$; less sweet, less soluble, hydrolysed to monosaccharides; energy source.

Q38. [SLO 9] Why is sucrose called transport sugar in plants?

Ans: Food is transported in plants as sucrose because it is very soluble and relatively unreactive chemically.

Q39. [SLO 9] Write the sources and roles of maltose and lactose.

Ans: Maltose: from starch and glycogen breakdown; found in germinating seeds, used in brewing. Lactose: milk sugar (3-8% in milk), by-product of cheese; energy for infants.

Q40. [SLO 9] What is a glycosidic bond?

Ans: Covalent bond between two ring sugars formed by condensation with release of one water; named by carbon numbers, e.g. alpha-1,4.

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Q41. [SLO 9] Name the monomers and linkage of sucrose, maltose and lactose.

Ans: Sucrose: alpha-glucose + beta-fructose, alpha-1,2. Maltose: two alpha-glucose, alpha-1,4. Lactose: beta-galactose + beta-glucose, beta-1,4.

Q42. [SLO 10] Differentiate between homo- and heteropolysaccharides.

Ans: Homopolysaccharides have one kind of monosaccharide (starch, glycogen, cellulose, chitin). Heteropolysaccharides have different kinds (agar, pectin, peptidoglycan).

Q43. [SLO 10] Why are polysaccharides good storage molecules?

Ans: They are large and insoluble (no osmotic effect), fold compactly and are easily hydrolysed to sugars when needed.

Q44. [SLO 10] Differentiate between amylose and amylopectin.

Ans: Amylose: unbranched, alpha-1,4, soluble in hot water. Amylopectin: branched with alpha-1,6, insoluble.

Q45. [SLO 10] Why is glycogen called animal starch?

Ans: It is the storage carbohydrate of animals (liver and muscles), made of alpha-glucose like starch but more branched.

Q46. [SLO 10] Why can humans not digest cellulose? What is its use in the diet?

Ans: Humans lack cellulase enzyme. As roughage it prevents abnormal absorption of food in the intestine.

Q47. [SLO 10] What is chitin? Where is it found?

Ans: A homopolysaccharide of N-acetyl glucosamine with beta-1,4 bonds, in fungal cell walls (fungal cellulose) and arthropod exoskeleton.

Q48. [SLO 10] How is starch detected? What colour does glycogen give?

Ans: Iodine test: starch gives dark blue-black, glycogen gives red colour.

Q49. [SLO 11] Define protein, amino acid and essential amino acid.

Ans: Protein: polymer of amino acids. Amino acid: building block with alpha C, H, $-NH_2$, -COOH and R group. Essential amino acid: cannot be made by body, must come from diet.

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Q50. [SLO 11] Write the structural formula of an amino acid.

Ans: H_2N - CH(R) - COOH: central alpha carbon with H, amino group, carboxyl group and R group.

Q51. [SLO 11] Name the nine essential amino acids.

Ans: Lysine, leucine, isoleucine, histidine, methionine, phenylalanine, threonine, valine and tryptophan.

Q52. [SLO 12] What is a peptide bond? How is it formed and broken?

Ans: C-N bond between -COOH of one amino acid and -NH_2 of another formed by condensation (water lost); broken by hydrolysis with enzymes like pepsin and trypsin.

Q53. [SLO 12] Define primary, secondary, tertiary and quaternary structure.

Ans: Primary: linear amino acid sequence. Secondary: alpha-helix / beta-pleated sheet by H-bonds. Tertiary: globular folding by ionic bonds and disulphide bridges. Quaternary: several globules joined by hydrophobic interactions.

Q54. [SLO 13] What is the significance of amino acid sequence?

Ans: Sequence decides the function of the protein; a change due to point mutation can cause serious defects as in sickle cell anaemia.

Q55. [SLO 13] What causes sickle cell anaemia?

Ans: A point mutation in the beta-globin gene replaces glutamic acid by valine at position 6 of the beta chain of haemoglobin; RBCs become sickle shaped and block vessels.

Q56. [SLO 14] Differentiate between fibrous and globular proteins.

Ans: Fibrous: filament shaped, insoluble, elastic, secondary structure (collagen, keratin). Globular: spherical, soluble, tertiary/quaternary (enzymes, hormones).

Q57. [SLO 14] Name three structural proteins and three functional proteins.

Ans: Structural: collagen, elastin, keratin. Functional: enzymes, hormones, antibodies (also haemoglobin).

Q58. [SLO 14] What is the biuret test?

Ans: To 2 ml protein solution add NaOH and a few drops of 1% $CuSO_4$; pink / violet colour shows protein.

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Q59. [SLO 15] Define lipids. Why do they give more energy than carbohydrates?

Ans: Heterogeneous organic compounds insoluble in water, soluble in organic solvents. They have much C and H but less oxygen, so give about double the energy.

Q60. [SLO 15] What are the roles of lipids?

Ans: Membrane components, energy store, steroid hormones, protection, waterproofing, insulation, buoyancy.

Q61. [SLO 16] Write any four properties of acylglycerols.

Ans: Less dense than water; insoluble in water; saturated long chains have higher melting point (tristearin 71 C, triolein -17 C); emulsifying power; hydrolysed to fatty acids and glycerol.

Q62. [SLO 16] Why are phospholipids called amphipathic?

Ans: They have both a polar hydrophilic head and non-polar hydrophobic tails.

Q63. [SLO 16] Write the roles of waxes and terpenes.

Ans: Waxes: polishing, waterproofing, candles, cosmetics, paints. Terpenes: anti-inflammatory, antioxidant, anticancer, antibiotic, antidiabetic.

Q64. [SLO 17] Define esterification and acylglycerol.

Ans: Esterification: reaction of an alcohol with an acid releasing water. Acylglycerol: ester of glycerol and fatty acids.

Q65. [SLO 17] Why are triacylglycerols called neutral lipids?

Ans: All three -OH groups of glycerol are occupied by fatty acids, so no charge-bearing -OH group remains.

Q66. [SLO 17] Differentiate between saturated and unsaturated fatty acids.

Ans: Saturated: no double bond, solid at room temperature, animal fats (palmitic). Unsaturated: one or more double bonds, liquid, plant oils (oleic).

Q67. [SLO 17] What is lecithin? What is an isoprene unit?

Ans: Lecithin = phosphatidylcholine, commonest phospholipid. Isoprene is the five-carbon building block of terpenes.

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Q68. [SLO 18] Write the structure and examples of steroids.

Ans: Nucleus of 17 carbons in four rings (three 6-carbon, one 5-carbon). Examples: cholesterol, testosterone, progesterone, estrogen, bile acids, vitamin D.

Q69. [SLO 18] What are prostaglandins? How does aspirin act?

Ans: Local hormones derived from arachidonic acid causing fever, inflammation and pain, and regulating platelets. Aspirin inhibits their synthesis, reducing fever and pain.

Q70. [SLO 19] Define nucleotide and nucleoside.

Ans: Nucleotide: pentose sugar + phosphate + nitrogenous base. Nucleoside: base + pentose sugar only.

Q71. [SLO 19] Differentiate between purines and pyrimidines.

Ans: Purines (adenine, guanine) have double ring; pyrimidines (cytosine, thymine, uracil) have single ring. Thymine only in DNA, uracil only in RNA.

Q72. [SLO 19] Why is the ribose sugar absent in DNA / what sugar does DNA have?

Ans: DNA has deoxyribose (H at 2' carbon); RNA has ribose (OH at 2').

Q73. [SLO 20] Why is ATP called energy currency of the cell?

Ans: Hydrolysis of its high energy phosphate bonds to ADP and Pi releases energy for cell activities, and it is re-made again.

Q74. [SLO 20] What is NAD?

Ans: Nicotinamide adenine dinucleotide: two nucleotides joined by their phosphates; works as a coenzyme.

Q75. [SLO 21] What is a phosphodiester bond?

Ans: Bond between 5' phosphate of one nucleotide and 3' -OH of the next, formed by condensation releasing pyrophosphate.

Q76. [SLO 22] State Chargaff's rule.

Ans: In DNA, adenine = thymine and guanine = cytosine, so purines : pyrimidines is 1 : 1.

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Q77. [SLO 22] What are the dimensions of the DNA double helix?

Ans: Width 2 nm, one turn 3.4 nm, 10 base pairs per turn.

Q78. [SLO 22] What is meant by antiparallel strands of DNA?

Ans: The two chains run in opposite directions: one 5' to 3' downward, the other 5' to 3' upward.

Q79. [SLO 23] Differentiate between mRNA, rRNA and tRNA.

Ans: mRNA carries the genetic message (codons); rRNA forms ribosome (80% of RNA); tRNA is smallest, cloverleaf, carries an amino acid and has an anticodon.

Q80. [SLO 23] What are codon and anticodon?

Ans: Codon: triplet of nucleotides in mRNA coding for one amino acid. Anticodon: three bases on tRNA complementary to the codon.

Q81. [SLO 24] State the central dogma of life.

Ans: Flow of genetic information from DNA to RNA (transcription) and from RNA to protein (translation).

Q82. [SLO 24] What is gene expression?

Ans: Transcription and translation together are called gene expression.

Q83. [SLO 25] What are conjugated molecules? Give examples.

Ans: Molecules joined with other kinds of molecules: glycolipids, glycoproteins, lipoproteins, nucleoproteins.

Q84. [SLO 25] Write the roles of glycoproteins and lipoproteins.

Ans: Glycoproteins: hormones, transport proteins, receptors, blood group antigens. Lipoproteins: found in milk, blood, nucleus, egg yolk and chloroplasts.

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Chapter 2: Long Questions with Answers (SLO based)

Q1. [SLO 2] Describe briefly the different types of bonds and interactions found in biological molecules.

Ans:

Type	Description
Covalent bond	Sharing of electrons; most prevalent; peptide, glycosidic, phosphodiester, disulphide, ester bonds.
Ionic bond	Electrostatic attraction between + and - ions; salts, some amino acids, phosphate of DNA, enzyme-substrate.
Hydrogen bond	H on O/N/F attracted to lone pair of another electronegative atom; DNA base pairs, protein folding.
Hydrophobic interaction	Non-polar parts cluster away from water; membranes, protein folding.
Hydrophilic interaction	Polar groups interact with water; hydration, solubility of proteins, DNA, enzymes.

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Q2. [SLO 4] Describe condensation and hydrolysis reactions for making and breaking of macromolecules. Draw sketches.

Ans:

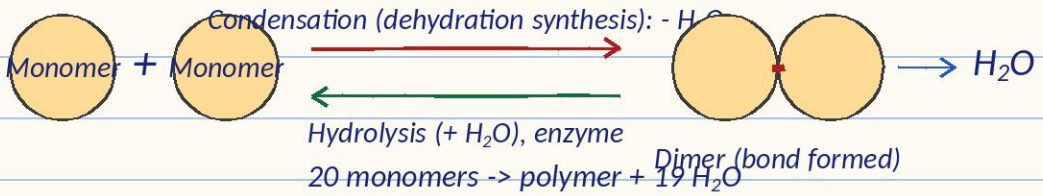

Fig: Condensation and hydrolysis

- **Condensation:** -OH from one monomer and -H from another are removed as water; a bond forms. Alpha-glucose + alpha-glucose = maltose (dimer);

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repetition gives amylose (polymer). n monomers release (n - 1) water. Needs enzyme and energy-rich monomers.

- ***Hydrolysis:*** *water splits the bond: -OH goes to one monomer and -H to the other; amylose to alpha-glucose by amylase. All digestion is hydrolysis.*

Q3. [SLO 5] Explain the properties of water that make it the medium of life.

Ans:

- ***High polarity:*** *universal solvent; reactions in aqueous medium.*
- ***Hydrogen bonding:*** *liquid at temperatures of life; cohesion and adhesion give transport.*
- ***High specific heat:*** *heat used to break H-bonds; temperature stabilizer.*
- ***High heat of vapourization (574 cal/g):*** *sweating cools the body.*
- ***Hydrophobic exclusion:*** *keeps lipid bilayer intact.*
- ***Ionization:*** *H^+ and OH^- affect pH and reactions.*
- ***Lower density of ice:*** *ice floats, insulates, aquatic life survives.*

Q4. [SLO 6-7] Define carbohydrates and classify them. Compare the properties of monosaccharides, oligosaccharides and polysaccharides.

Ans:

- *Carbohydrates: polyhydroxy aldehydes / ketones or derivatives yielding them on hydrolysis; general formula $C_n(H_2O)_n$.*

Monosaccharides	***Oligosaccharides***	***Polysaccharides***
1 unit	*2-10 units*	*>10 units*
Not hydrolysable	*Yield 2-10 monosaccharides*	*Yield at least 11*
Highly soluble	*Less soluble*	*Insoluble*
Sweetest	*Less sweet*	*Tasteless*
Glucose, fructose, ribose	*Sucrose, maltose, lactose*	*Starch, glycogen, cellulose, chitin*

- *Monosaccharides also classified: aldoses / ketoses; trioses to heptoses (3C-7C).*

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Q5. [SLO 7] Compare and contrast the properties and roles of monosaccharides. Write their formulae.

Ans:

- General formula $(CH_2O)_n$; glucose $C_6H_{12}O_6$; ribose $C_5H_{10}O_5$; deoxyribose $C_5H_{10}O_4$.
- Properties: cannot be hydrolysed; highly soluble (H-bonds with water); sweet; reducing sugars; stereoisomerism; open chain or ring (furanose / pyranose).

Monosaccharide	Role
Glyceraldehyde, dihydroxyacetone (3C)	Intermediates in photosynthesis and respiration.
Ribose, deoxyribose (5C)	Components of RNA and DNA nucleotides.
Ribulose (5C)	Intermediate in photosynthesis.
Glucose (6C)	Main respiratory substrate.
Fructose (6C)	Intermediate in respiration; sweet fruit sugar.
Galactose (6C)	Component of milk sugar lactose.

Q6. [SLO 8] Compare the isomers and stereoisomers of glucose.

Ans:

- Isomers: same formula $C_6H_{12}O_6$; glucose (aldo) and fructose (keto) are functional group isomers.
- Glucose has 4 asymmetric carbons so $2^4 = 16$ stereoisomers.

Glucopyranose ring: O links C1 and C5 (pyranose = 6-membered)

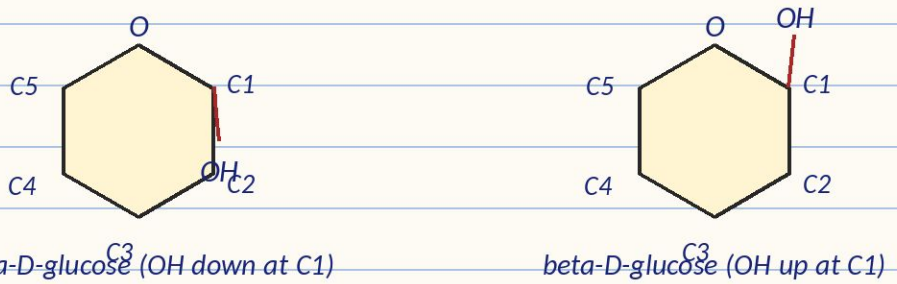

Fig: Anomers of glucose

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Stereoisomer	Difference
Anomers (alpha, beta)	At C1 only; -OH down = alpha, up = beta.
Enantiomers (D, L)	Mirror images; -OH on C5 right (D) or left (L).
Diastereoisomers	Differ at more than one asymmetric C; not mirror images (D-glucose, D-altrose).
Epimers	Differ at one asymmetric C (D-glucose, D-mannose).

Q7. [SLO 9] Distinguish the properties and roles of disaccharides. Describe glycosidic bonds in sucrose, maltose and lactose.

Ans:

- Disaccharides: $C_{12}H_{22}O_{11}$, less sweet and soluble, hydrolysed to monosaccharides, energy source.

Disaccharide	Formation and role
Sucrose	alpha-glucose C1 -OH + beta-fructose C2 -H: alpha-1,2 (alpha-1, beta-2) linkage. Transport sugar of plants; cane sugar.
Maltose	Two alpha-glucose: C1 -OH with H of C4 -OH: alpha-1,4 linkage. From starch breakdown; germinating seeds.
Lactose	beta-galactose C1 -OH with H of C4 of beta-glucose: beta-1,4 linkage. Milk sugar; energy for infants.

- Each bond forms by condensation, releasing one water molecule.

Q8. [SLO 10] Describe the structure, properties and roles of starch, glycogen, cellulose and chitin.

Ans:

Polysaccharide	Structure and role
Starch	alpha-glucose; amylose (alpha-1,4, unbranched) + amylopectin (alpha-1,6 branches). Plant storage in root, stem, seeds. Iodine: blue-black.
Glycogen	alpha-glucose, highly branched; animal storage in liver and muscles. Iodine: red.

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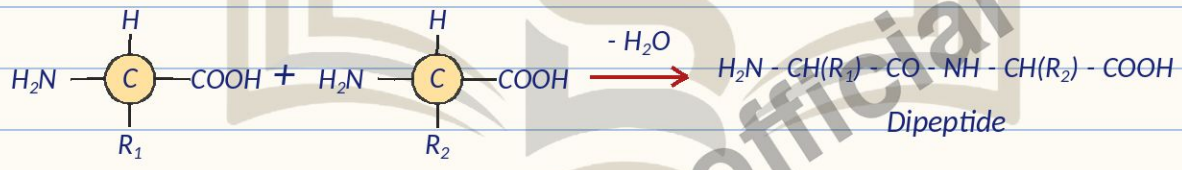
Polysaccharide	Structure and role
Cellulose	beta-glucose, beta-1,4, unbranched; plant cell wall; no iodine colour; not digestible by humans (roughage).
Chitin	N-acetyl glucosamine, beta-1,4; fungal cell wall and arthropod exoskeleton; fungal cellulose.

Q9. [SLO 11-12] Define protein and amino acid. Describe the synthesis and breakage of peptide linkages.

Ans:

- Protein: polymer of amino acids; amino acid: alpha C + H + $-NH_2$ + -COOH + R group (20 common).



CO - NH = peptide bond (C of -COOH to N of $-NH_2$)

General amino acid: alpha C + H + amino ($-NH_2$) + carboxyl (-COOH) + R group

Fig: Amino acids and formation of peptide bond

- Synthesis (condensation): on ribosome during translation; -OH of carboxyl and -H of amino group form water; C-N peptide bond gives dipeptide with N-terminal and C-terminal ends; new amino acid joins at C-end to give polypeptide.
- Breakage (hydrolysis): water added; in gut by pepsin, trypsin, erepsin; in cell by lysosomal enzymes to release amino acids.

Q10. [SLO 11] Write notes on essential and non-essential amino acids.

Ans:

- **Essential (9):** body cannot make; from diet (milk, egg, fish, meat, lentils): lysine, leucine, isoleucine, histidine, methionine, phenylalanine, threonine, valine, tryptophan. Build and repair muscle and bone; precursors of hormones

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and neurotransmitters; deficiency causes health problems.

- ***Non-essential (11):*** *made by body from other amino acids / food: alanine, arginine, asparagine, aspartic acid, cysteine, glutamic acid, glutamine, glycine, proline, serine, tyrosine. Help detoxification, RBC and WBC synthesis, brain and heart function.*

Q11. [SLO 12] Describe the levels of organization of protein structure.

Ans:

Level	*Details*
Primary	*Linear sequence of amino acids by peptide bonds; made on ribosome.*
Secondary	*Alpha-helix or beta-pleated sheet by H-bonds.*
Tertiary	*Further folding into globular shape by ionic bonds and disulphide bridges.*
Quaternary	*Two or more polypeptide globules joined by hydrophobic interaction (e.g. haemoglobin).*

Q12. [SLO 13] Justify the significance of the sequence of amino acids through the example of sickle cell haemoglobin.

Ans:

- *Sequence of amino acids is the primary structure, set by the nucleotide sequence of DNA, and it determines how a protein works.*
- *Normal haemoglobin: 2 alpha chains (141 amino acids) + 2 beta chains (146); carries oxygen; RBCs disc shaped.*
- *Sickle cell anaemia: point mutation in beta-globin gene (one nucleotide replaced) changes glutamic acid to **valine at position 6** of the beta chain (HbS).*
- *Result: sickle (crescent) shaped, stiff, sticky RBCs that block blood flow in limbs and organs, causing pain and organ damage.*

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Q13. [SLO 14] Classify proteins as globular and fibrous. List the roles of structural and functional proteins with three examples each.

Ans:

Fibrous	***Globular***
Filament shaped	*Spherical*
Secondary structure	*Tertiary / quaternary*
Insoluble, elastic, not crystallized	*Soluble, inelastic, crystallized*
Collagen, keratin, actin, myosin	*Enzymes, hormones, antibodies*

- ***Structural:*** *collagen (bone, cartilage, tendons), elastin (connective tissue), keratin (hair, nails, feathers, horns), histones (DNA into chromosome).*
- ***Functional:*** *enzymes (speed up reactions), hormones (regulate glucose, calcium), antibodies (immunity), haemoglobin (oxygen transport), fibrinogen (clotting), casein / ovalbumin (storage).*

Q14. [SLO 15-16] Define lipids. Describe the properties and roles of acylglycerols, phospholipids, terpenes and waxes.

Ans:

- *Lipids: hydrophobic organic compounds (fats, oils, waxes, steroids), soluble in organic solvents; double energy of carbohydrates.*

Lipid	***Properties and roles***
Acylglycerols	*Less dense than water, insoluble in water; saturated long chain = higher melting point; hydrolysis to fatty acid + glycerol; main fat store; excess gives obesity, diabetes; hydrogenation gives margarine.*
Phospholipids	*Amphipathic; fix proteins in bilayer; regulate permeability; fat absorption; transport cholesterol; lung surfactant.*
Terpenes	*Anti-inflammatory, analgesic, antioxidant, anticancer, antibiotic, antidiabetic.*
Waxes	*Waterproofing, polishing; candles, cosmetics, crayons, paints, tyres.*

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Q15. [SLO 17] Illustrate the molecular structure (making and breaking) of an acylglycerol, a phospholipid and a terpene.

Ans:

- ***Acylglycerol:*** *glycerol + fatty acid (R-COOH) = ester bond + water (esterification). 1, 2 or 3 fatty acids = mono-, di-, triacylglycerol. Hydrolysis (heat, aqueous acid) breaks it into glycerol and fatty acids.*
- ***Phospholipid:*** *glycerol + 2 fatty acids + phosphate (phosphatidic acid), joined to choline / ethanolamine / inositol / serine. Lecithin = with choline.*

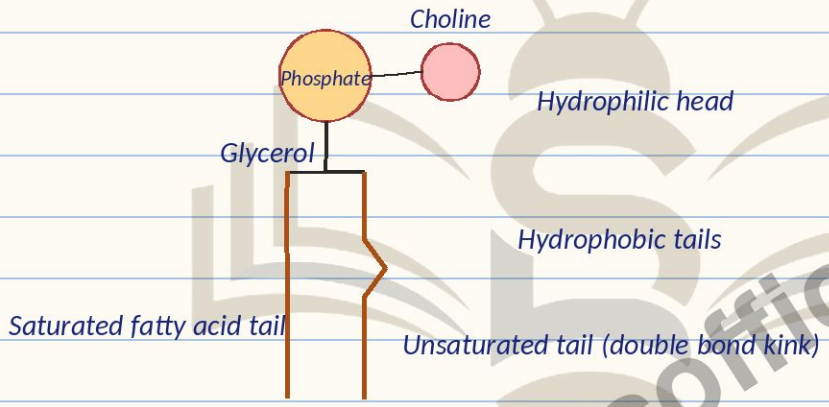

Fig: Structure of a phospholipid (phosphatidylcholine)

- ***Terpene:*** *five-carbon isoprene units condense: 2 = monoterpene (menthol), 4 = diterpene (vitamin A, phytol), 6 = triterpene (ambrein), many = natural rubber.*

Q16. [SLO 17] Write the chemical structure of a single phospholipid (glycerol, phosphate group, one unsaturated and one saturated fatty acid tail).

Ans:

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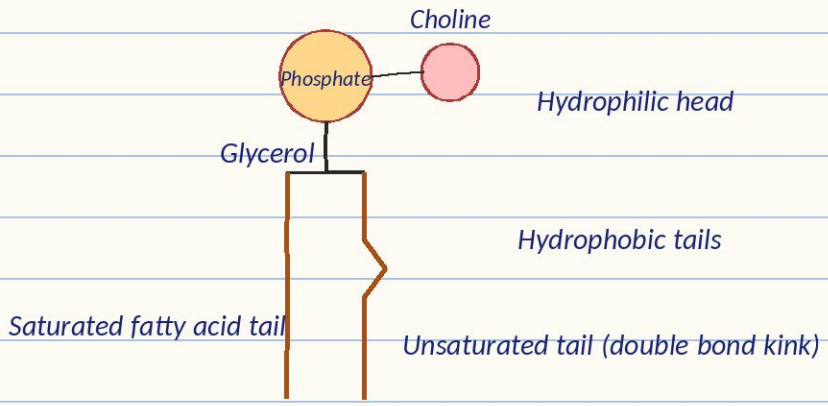

Fig: Structure of a phospholipid (phosphatidylcholine)

- *Glycerol (3 carbon backbone): first -OH esterified with one saturated fatty acid (straight tail), second -OH with one unsaturated fatty acid (kinked tail due to double bond), third -OH with phosphate, which carries an organic group such as choline.*
- *Head (phosphate + group) is polar / hydrophilic; tails are non-polar / hydrophobic.*

Q17. [SLO 18] Evaluate steroids and prostaglandins as important groups of lipids.

Ans:

- ***Steroids:** 4 fused rings (17 carbons); made from isoprene units; cholesterol is membrane component and precursor of bile acids (emulsify fats), testosterone, progesterone and estrogen; vitamin D regulates calcium metabolism.*
- ***Prostaglandins:** local hormones from arachidonic acid in nearly all mammalian tissues; lower or raise blood pressure; cause fever, inflammation (redness, heat, pain, swelling) and pain; regulate platelet aggregation / blood clotting.*
- *Aspirin inhibits prostaglandin synthesis and so reduces fever and pain.*

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Q18. [SLO 19-20] Describe nucleic acids and the molecular structure of nucleotides. Give examples of a mononucleotide (ATP) and a dinucleotide (NAD).

Ans:

- *Nucleic acids (DNA and RNA): unbranched polymers of nucleotides. Nucleotide = pentose sugar + phosphate + nitrogenous base. Base on 1' C gives nucleoside; phosphate on 5' C gives nucleotide (mono-, di-, triphosphate).*
- *Bases: pyrimidines C, T (DNA only), U (RNA only); purines A, G.*
- ***ATP** (mononucleotide): adenine + ribose + 3 phosphates; high-energy bonds; hydrolysis to ADP + Pi gives energy: energy currency.*
- ***NAD** (dinucleotide): nicotinamide-sugar-phosphate joined to adenine-sugar-phosphate; coenzyme.*

Q19. [SLO 21] Illustrate the formation of phosphodiester bond.

Ans:

- *Nucleoside triphosphate (NTP) with 5' phosphate is held close to the free 3' -OH of the last nucleotide of the growing chain.*
- *A phosphodiester bond forms between them; the bond between alpha and beta phosphates breaks and pyrophosphate is released.*
- *Hydrolysis of pyrophosphate releases energy and drives polymerization.*
- *Polynucleotide has free 5' phosphate at one end and free 3' -OH at the other; read 5' to 3'.*

Q20. [SLO 22] Explain the double helical structure of DNA as proposed by Watson and Crick.

Ans:

- *Basis: Chargaff's rule (A = T, G = C) and X-ray diffraction of Wilkins and Franklin (double helix, 2 nm width, 3.4 nm turn). Model proposed in 1953 at Cambridge.*

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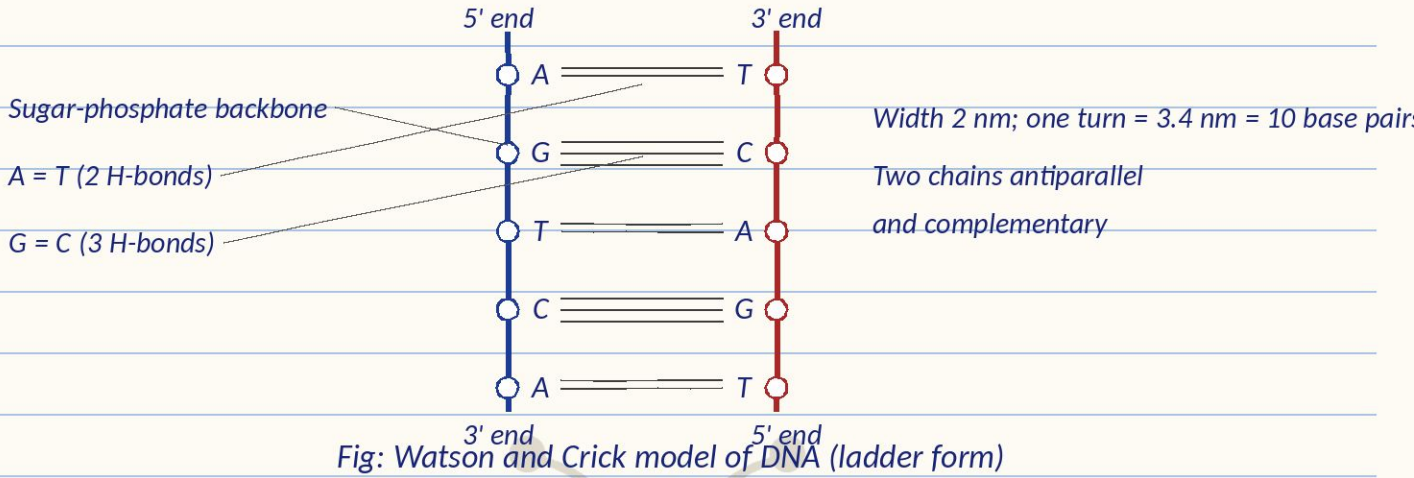

Fig: Watson and Crick model of DNA (ladder form)

- *Two polynucleotide chains, antiparallel (5' to 3' in opposite directions), constant width 2 nm.*
- *Specific base pairing: A with T by 2 H-bonds, G with C by 3 H-bonds; chains complementary; 10 base pairs per turn.*
- *Any sequence along a strand is possible; sequence is specific for species and individuals.*

Q21. [SLO 23] Explain the general structure of RNA. Distinguish the three types of RNA in terms of functions.

Ans:

- *RNA: polymer of ribonucleotides (ribose, U instead of T); generally single stranded; some regions double stranded; made from DNA template.*

Type	Structure and function
mRNA	3-4% of RNA; codons; carries message from DNA to ribosome.
rRNA	80% of RNA; largest; makes ribosome; platform for protein synthesis.
tRNA	Smallest (75-90 nt); cloverleaf, 3 loops; anticodon; 3' CCA end carries amino acid; about 45 kinds in humans.

Q22. [SLO 24] Discuss the central dogma of life.

Ans:

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Transcription + translation = gene expression

Fig: Central dogma of life

- *Genes hold the information; information flows DNA to RNA to protein.*
- ***Transcription:** DNA strands of a gene unwind and one strand is copied into single-stranded mRNA with complementary RNA nucleotides (A-U, C-G); DNA rewinds.*
- ***Translation:** mRNA binds ribosome in cytoplasm; tRNAs with anticodons carry amino acids; each codon = one amino acid; peptide bonds make polypeptide.*
- *Transcription + translation = gene expression.*

Q23. [SLO 25] Define conjugated molecules and describe the roles of glycolipids, glycoproteins, lipoproteins and nucleoproteins.

Ans:

Molecule	***Composition and role***
Glycolipids	*Lipid + sugars; brain white matter, myelin sheath, chloroplast membrane.*
Glycoproteins	*Protein + carbohydrate; hormones, transport, structure, receptors; blood group antigens.*
Lipoproteins	*Protein + phospholipid; in milk, blood, nucleus, egg yolk, chloroplasts.*
Nucleoproteins	*Basic protein + nucleic acid; chromosomes and ribosomes.*

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